

Clinical *Campylobacter jejuni* isolates: genomes and genetic tools

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25 Abstract

26 *Campylobacter jejuni* is a major cause of food-borne gastroenteritis and is responsible for
 27 substantial mortality and economic losses in meat and dairy production. Detecting *C. jejuni* in
 28 contaminated food samples remains difficult because current assays are culture-based, slow, and
 29 can yield false positives. As a result, contamination may not be identified for several days, limiting
 30 detection at the point of production. Developing improved assays has also been challenging
 31 because *Campylobacter* genetics and the biology of clinical isolates remain poorly understood.
 32 Here, we expand the *C. jejuni* genetic toolbox by sequencing two strains, HC1 and RM1164,
 33 derived from patient and food samples. We identified two cryptic plasmids in HC1, one potentially
 34 capable of conjugation and another conferring tetracycline resistance. We also engineered a
 35 mobilizable plasmid carrying an *OriT* sequence that can be transferred from *Escherichia coli*
 36 donor strains to *C. jejuni* RM1164 by conjugation. Together, these clinical isolates and the plasmid
 37 system expand the genetic tools available for *C. jejuni*.

38

39 **Key words:** *Campylobacter jejuni*, conjugation, antibiotic resistance, whole-genome assembly

40 Introduction

41 Food-borne illnesses are a growing public health concern, with cases expected to rise across
 42 North America [1,2]. Millions of infections are reported each year, leading to substantial mortality
 43 and billions of dollars in food recall costs [1,3]. Pathogenic bacteria are the main causes of food-
 44 borne disease, and *Campylobacter jejuni* is consistently ranked as one of the most common
 45 bacterial causes of gastroenteritis [4]. Infections are often linked to raw or undercooked poultry,
 46 meat, or other contaminated dairy produce [5]. In addition to acute gastroenteritis,
 47 campylobacteriosis can trigger severe autoimmune sequelae, including Guillain–Barré and

48 Miller–Fisher syndromes [6,7]. Cases of campylobacteriosis are expected to rise in coming years,
49 as climate change may promote conditions favourable for *Campylobacter* growth and extend its
50 seasonal prevalence in avian hosts [1,8]. Together, these factors highlight the importance of
51 improving *Campylobacter* detection in contaminated food products and reducing its public health
52 burden.

53 The average *C. jejuni* genome is between 1.61 and 1.87 Mb, with a GC content of ~30.9%
54 and 1600-1800 protein coding genes [9,10]. Comparative genomics indicates a conserved core
55 genome comprised of approximately 1400-1500 core genes [11]. Some *C. jejuni* strains also
56 harbour cryptic plasmids that carry antimicrobial resistance genes and can be self-transmissible,
57 such as pTet and pVir, which span 1.3–208 kb and encode distinct replicon types and relaxases
58 that may contribute to strain-specific pathogenicity in *C. jejuni* [12]. Type III (T3SS) and type VI
59 (T6SS) secretion systems have been described in *C. jejuni*, with about 16–20% of sequenced
60 strains carrying T6SS loci associated with adhesion and invasion. Strains that encode T3SS
61 express *Campylobacter* invasion antigens that promote uptake of *C. jejuni* into host cells [13].

62 A genetic toolbox for *C. jejuni* is currently under development, containing several methods
63 for DNA deliver, including natural transformation, electroporation, and conjugation [14]. Natural
64 transformation into *C. jejuni* can be achieved by spreading a layer of culture across an agar plate
65 and adding an additional layer of transformant DNA carrying a selection marker [15].
66 Electroporation into *C. jejuni* has also been previously documented, including protocols and
67 settings for DNA transfer and producing electrocompetent *C. jejuni* cells [16]. For conjugation,
68 the mobilizable plasmid pRK212.1 has been used to deliver replicating vectors into *C. jejuni* [14]
69 .Other shuttle vectors have also been developed that carry *Campylobacter* and *E. coli* replicons,
70 multiple cloning sites, and selectable markers such as kanamycin and ampicillin [17]. There are
71 also methods for genome mutagenesis that use Tn5, EZ:Tn5, and mariner transposons [18–20].
72 Together, these tools enable gene knockouts, chromosomal insertions, random mutagenesis,

73 and recombinant gene expression in *C. jejuni*. However, we and others have found that many of
 74 these approaches perform inconsistently, with transformation efficiencies varying widely
 75 between *C. jejuni* strains [21,22]. Most of these tools were developed for commonly used
 76 reference strains, including NCTC 11168, and have not been well documented in more virulent
 77 isolates such as *C. jejuni* 81-17623 [23]. This highlights the need for *C. jejuni* genetic tools that
 78 work more broadly, especially in clinically relevant isolates.

79 Here, we build on the existing *Campylobacter* genetic toolbox by introducing two new *C.*
 80 *jejuni* strains, RM1164 and HC1, isolated from contaminated food and clinical samples. We
 81 sequenced both strains to identify genomic features of interest and to assess the presence of
 82 cryptic plasmids. Two cryptic plasmids were identified in *C. jejuni* strain HC1 that appear to carry
 83 conjugation machinery, virulence factors, and genes conferring resistance to tetracycline. To host
 84 recombinant genes in *C. jejuni*, we developed a shuttle plasmid (pCJ01) capable of replication in
 85 *E. coli* and *C. jejuni*. To facilitate genetic manipulation of *C. jejuni*, we developed a shuttle plasmid
 86 (pCJ01) capable of replication in both *E. coli* and *C. jejuni*. pCJ01 also carries an origin of transfer,
 87 allowing its mobilization from a non-pathogenic *E. coli* donor strain into *C. jejuni*. Together, this
 88 work expands the *Campylobacter* genetic toolbox by introducing two diverse *C. jejuni* isolates
 89 and demonstrating plasmid transfer by conjugation.

90 **Materials and methods**

91 **Microbial strains and growth conditions**

92 *Campylobacter jejuni* strain RM1164 was provided by Dr. Michael Rieder. *C. jejuni* strain Human
 93 Clinical Isolate 1 (HC1) was provided by London Health Sciences Centre (LHSC). *C. jejuni* strains
 94 were grown in a microaerobic atmosphere (85% N₂, 10% CO₂, and 5% O₂) at 37°C on Columbia
 95 Blood Agar Base (CBA) (Difco™ 279240) supplemented with 5% sheep blood (Cedarlane

Laboratories Ltd., Hornby, ON, Canada), 10 $\mu\text{g/mL}$ vancomycin, and 5 $\mu\text{g/mL}$ trimethoprim as background antibiotics. Growth media was supplemented with 30 $\mu\text{g/mL}$ kanamycin as required.

Saccharomyces cerevisiae strain VL6-48 (American Type Culture Collection [ATCC], catalog number: MYA-3666) was grown in 2X YPAD media with 1.5% agar (w/v) and 1 M D-sorbitol. Yeast cultures were incubated at 30°C, with liquid cultures placed on a shaker set to 225 rpm. Agar plates were sealed with parafilm to reduce desiccation. Yeast media were prepared as previously described [24].

Escherichia coli (EPI300, Epicenter) was grown at 37°C in Luria Broth (LB) supplemented with antibiotics as needed (50 $\mu\text{g/mL}$ kanamycin). *E. coli* ECGE101 (ΔdapA) [25], was grown at 37°C in LB supplemented with 60 $\mu\text{g/mL}$ diaminopimelic acid (DAP), and the appropriate antibiotics (50 $\mu\text{g/mL}$ kanamycin and/or 40 $\mu\text{g/mL}$ gentamycin). Cultures were placed in an incubator set to 37°C, with liquid cultures shaking at 225 rpm.

Whole-genome sequencing and phylogenetic analysis

Cultures of *C. jejuni* RM1164 and HC1 were grown to saturation, and genomic DNA was extracted using an alkaline lysis protocol as previously described [26]. Whole-genome sequencing was performed by Flow Genomics using their Bacterial Genomes service. Genomic DNA was sequenced on the Oxford Nanopore MinION, with a R10.4.1 flow cell and the SQK-RBK114.96 rapid barcoding kit following manufacturer's instructions (Oxford Nanopore Technologies). Basecalling was performed on Dorado (v0.9.6), read filtering with Filtlong (v0.3.1), genome assembly with Flye (v2.9.6), polishing with Medaka (v2.2.0), and genome annotation with either Bakta (v1.12) or pLannotate (v1.2.4). Publicly available *C. jejuni* genomes were downloaded from the National Centre for Biotechnology Information (NCBI), and after trimming redundant copies, we were left with 526 unique genomes. In addition, *C. jejuni* genomes of interest (NCTC 11168 and 81-176) were also isolated and included in downstream analyses. Mash (v2.3) was used to

120 generate sketches and distance matrices, and a custom python script converted the matrices into
121 phylogenetic trees. The tree was visualized using the Interactive Tree of Life [27].

122 **DNA isolation**

123 DNA was isolated from *C. jejuni* and *S. cerevisiae* using a modified alkaline lysis protocol as
124 previously described [26]. We modified step 1 for *C. jejuni* DNA isolation, which is described
125 below. Step 1 was modified for *C. jejuni* to isolate pCJ01 with reduced contamination and at
126 higher concentrations. Step 1: *C. jejuni* transformants that were re-patched twice were grown to
127 saturation on Columbia Blood agar supplemented with sheep blood and the appropriate
128 antibiotics (10 μ g/mL vancomycin and 5 μ g/mL trimethoprim as background antibiotics, with 30
129 μ g/mL kanamycin for the transformants). Colonies were passaged as large streaks to ensure
130 there were sufficient cells for DNA isolation; typically, a single plate would be used to passage a
131 single colony. Plates were grown over 2-3 days. Cells were then scraped and resuspended in
132 250 μ L resuspension buffer. For *E. coli*, we used the EZ-10 DNA Miniprep Kit to isolate and purify
133 DNA (BioBasic Inc. BS414). DNA was eluted in 30 μ L ddH₂O.

134 **PCR amplification**

135 All primers are listed in Supplementary S1 Table. Fragments for assembly were amplified using
136 PrimeSTAR GXL DNA Polymerase (Takara, R050A) using the PCR protocol. PCR reactions were
137 run over 30-35 cycles depending on amplification efficiency. A DpnI (New England Biolabs,
138 R0176) was used to removed template DNA from the amplifications, following manufacturer's
139 protocols. Fragments were run over a 1% agarose gel and purified using the PureLink PCR
140 Purification Kit (Invitrogen, K310001) and eluted using ddH₂O.

141 **Screening *C. jejuni* and *E. coli* transformants**

142 Five individual transformant colonies were passaged three times on solid media with the
143 appropriate antibiotic, and DNA was isolated and PCR screened for the inserted kanamycin
144 resistance gene. 1 μ L of DNA isolated by alkaline lysis was used as a template in a 20 μ L
145 SuperPlex PCR reaction (Takara, 638543). Reactions were performed according to
146 manufacturer's protocol for 30 cycles.

147 **Plasmid construction**

148 The pCJ01 plasmid was developed from the pHflu3 plasmid [28] by the addition of the *C. jejuni*
149 origin of replication (*OriC*) and kanamycin resistance gene using yeast assembly. pHflu3 was
150 linearized by XhoI (New England Biolabs, R0146). The *OriC* was amplified from *C. jejuni* RM1164
151 genomic DNA, and the kanamycin resistance gene was amplified from pMW10 [29] using primers
152 that added 40 bp of overlapping homology to each end (S1 Table). The resulting amplicons and
153 linearized pHflu3 were combined in an equimolar ratio and co-transformed into *S. cerevisiae*
154 spheroplasts, yielding pCJ01. Yeast assembly was conducted as previously described [24] using
155 mixtures of DNA fragments in place of a bacterial donor. Following assembly, DNA was isolated
156 from pooled *S. cerevisiae* transformants and electroporated into *E. coli* EPI300. Individual
157 colonies were screened for correct plasmid assembly by multiplex PCR and diagnostic restriction
158 digestion.

159 **Electroporation to *C. jejuni***

160 The preparation of electrocompetent cells was performed as previously described [14]. *C. jejuni*
161 was streaked onto CBA supplemented with 5% sheep blood, 10 μ g/mL vancomycin and 5 μ g/mL
162 trimethoprim from a frozen glycerol stock and grown for 24 hours at 37°C under microaerophilic
163 conditions. The culture was replated onto a new plate and incubated for an additional 16 hours

under the same growth conditions. Following this incubation, the colonies were scraped and resuspended in 2 mL of Brain Heart Infusion (BHI) (Difco™ 211059) liquid broth without antibiotics. Cells were centrifuged at 16,000xg for 5 min at 4°C. The supernatant was discarded, and cell pellets were resuspended in 2 mL of ice-cold wash buffer (63% (v/v) glycerol/37% (w/v) sucrose). The centrifugation and resuspensions were repeated an additional two times. After the final centrifugation, the supernatant was discarded, and the pelleted cells were resuspended in 600 μ L of ice-cold wash buffer. The cells were then divided into 50 μ L aliquots in sterile microcentrifuge tubes and stored at -80°C.

Electroporation into *C. jejuni* follows as a protocol as previously described [14], with minor modifications. 50 μ L of *C. jejuni* cells were combined with 20 μ g of pCJ01 in a microcentrifuge tube and incubated on ice for 2-3 minutes. The mix was then transferred to a prechilled 2 mm cuvette (VWR) and pulsed in a GenePulser Xcell (Bio-Rad) at 2.5 kV voltage, 200 Ω resistance, and 25 μ F capacitance, with a time constant of ~5 msec. Cells recovered in 100 μ L SOC media (5 mL 2M MgCl₂, 10 mL 250 mM KCl, 20 mL 1M glucose, 0.5 g/L NaCl, 5 g/L yeast extract, 20 g/L tryptone) on Sheep Blood agar with Columbia Base plate supplemented with 10 μ g/mL vancomycin and 5 μ g/mL trimethoprim at 37°C under microaerophilic conditions for 5 hours. After recovery, cells were harvested by resuspending in 1 mL BHI broth without antibiotics, followed by centrifugation for 2 min at 14,000xg and discarding the supernatant. Pellets were resuspended in 100 μ L of BHI broth and spread across a Sheep Blood agar with Columbia Base plate supplemented with 10 μ g/mL vancomycin, 5 μ g/mL trimethoprim, and 30 μ g/mL kanamycin at 37°C under microaerophilic conditions. Colony forming units could be visualized on plates after 2-3 days.

186 Transformation into *E. coli*

187 Isolated plasmid DNA was added to 50 μ L electrocompetent *E. coli* cells in a 1.5 mL tube. The
 188 mix was gently pipetted to mix and transferred to a pre-chilled 2 mm electrocuvette. The cuvette
 189 was pulsed at 2.5 kV voltage, 200 Ω resistance, and 25 μ F capacitance. Cells recovered in 1 mL
 190 SOC media at 37°C with shaking at 225 rpm for 1 hour. Cells were collected by centrifugation for
 191 2 min at 10,000xg. The supernatant was discarded, and pellets were resuspended in 100 μ L of
 192 liquid LB and spread across a 1.5% agar (w/v) LB plate supplemented with or without DAP 60
 193 μ g/mL and the appropriate antibiotics (50 μ g/mL kanamycin and/or 40 μ g/mL gentamycin). Plates
 194 were incubated at 37°C, and transformants could be visualized within 24 hours.

195 *C. jejuni* plasmid recovery and digests

196 Following transformation and plating, 10 *C. jejuni* transformants were passaged three times on
 197 solid media with the appropriate antibiotics. 5 colonies were harvested from the plates by
 198 resuspending in ddH₂O. Cells were pelleted and DNA was isolated by alkaline lysis.
 199 Approximately 1 μ L of isolated DNA (~300 ng) was used to transform into *E. coli* as previously
 200 described. For each transformation, plasmid DNA was isolated using the EZ-10 DNA Miniprep
 201 Kit (BioBasic Inc. BS414) according to manufacturer's instructions. Isolated pCJ01 was digested
 202 using 5 units of XbaI (New England Biolabs, R0145L) in a 20 μ L reaction at 37°C for 1 hour.
 203 Complete digestion was confirmed on a 1% agarose gel.

204 Conjugation assays from *E. coli* to *C. jejuni*

205 The *E. coli* ECGE101 Δ dapA donor strain [25] harbouring pTA-Mob [30] were grown at 37°C
 206 overnight in 5 mL of LB media supplemented with 60 μ g/mL DAP, 40 μ g/mL gentamycin (donor
 207 only), and 50 μ g/mL kanamycin. Saturated *E. coli* cultures were diluted 1:10 into 5 mL of fresh
 208 LB and grown to an OD₆₀₀ of 0.5. The *C. jejuni* recipient strain was streaked on a Columbia Blood

209 Agar (CBA) plate containing 5% sheep blood and the background antibiotics for 16-20 hours at
 210 37°C under microaerophilic conditions. Cells were then resuspended from the plates in 2 mL of
 211 Brain Heart Infusion (BHI) broth without antibiotics and diluted to an OD₆₀₀ of ~ 1 in 1 mL of BHI
 212 media. For the *E. coli* donor or control strain, 1 mL of culture was washed twice by centrifugation
 213 at 10,000xg for 5 min at 4°C, discarding the supernatant, and adding 1 mL BHI broth without
 214 antibiotics. The *E. coli* donor and control pellets were then resuspended with 1 mL of the *C. jejuni*
 215 suspension. The bacterial mix was spun down at 12,000xg for 5 min at 4°C, and resuspended in
 216 100 µL BHI broth without antibiotics. This resuspension was then spread onto CBA plates
 217 supplemented with 5% sheep blood and 60 µg/mL DAP. These plates were incubated at 37°C
 218 for 5 hours under microaerophilic conditions, and then cells were scraped off with 1 mL of BHI
 219 broth, and adjusted final volume of 1 mL in a microfuge tube. A 10-fold serial dilution was
 220 performed with BHI broth, and 100 µL of each dilution was plated on selective CBA agar (5%
 221 sheep blood, 10 µg/mL vancomycin, 10 µg/mL trimethoprim (TMP), and 30 µg/mL kanamycin).
 222 Plates were incubated at 37°C for 2-3 days under microaerophilic conditions.

223 Antibiotic susceptibility assays

224 *C. jejuni* was grown on CBA agar supplemented with 5% sheep blood and the appropriate
 225 antibiotic (10 µg/mL vancomycin, 5 µg/mL trimethoprim, and/or 30 µg/mL kanamycin). Cells were
 226 scraped and resuspended in 1 mL of BHI broth and diluted to an OD₆₀₀ of 0.5 before being serially
 227 diluted in CBA media to either a 10⁻⁴ or 10⁻⁵ dilution. 3 µL of each dilution was plated on media
 228 containing either 2 µg/mL chloramphenicol, 5 µg/mL gentamycin, 5 µg/mL neomycin, 1 µg/mL
 229 tetracycline, 5 µg/mL kanamycin, or on a non-selective control. Plates were incubated for 3 days
 230 at 37°C under microaerophilic conditions before images were taken.

Results

Whole-genome sequencing of *C. jejuni* strains

Two *C. jejuni* strains were collected from contaminated food and patient-derived samples. Both strains were subject to long-read whole-genome sequencing, revealing genomic characteristics expected from other *C. jejuni* strains [31,32]. Both genomes trend towards the smaller size for *C. jejuni*, with genome sizes of 1.69 Mb and 1.62 Mb for HC1 and RM1164, respectively (Fig 1A/B). Primer pairs designed from *C. jejuni* NCTC 11168 were used to genotype HC1 and RM1164, with expected sizes ranging from 165-650 bp (Fig 1C). Interestingly, sequencing also identified two cryptic plasmids from *C. jejuni* HC1, pTet and pHC1 (Fig 1D/E). pHC1 appears to carry elements of conjugative machinery and virulence factors, although the current annotation seems to suggest that this is an incomplete Type IV Secretion System (T4SS). pTet does not carry complementary conjugation machinery, though it does carry a gene conferring resistance to tetracycline, enabling its continual propagation. No contigs beyond genomic DNA were identified for *C. jejuni* RM1164. A cursory investigation into the evolutionary ancestry of RM1164 and HC1 reveal that HC1 shares a fairly recent common ancestor with the first sequenced *C. jejuni* genome, the flagship strain NCTC 11168 (Fig 1F) [33]. However, RM1164 appears distant, sharing a recent common ancestor with neither HC1, NCTC 11168, nor *C. jejuni* 81-176, another highly virulent isolate of interest [34].

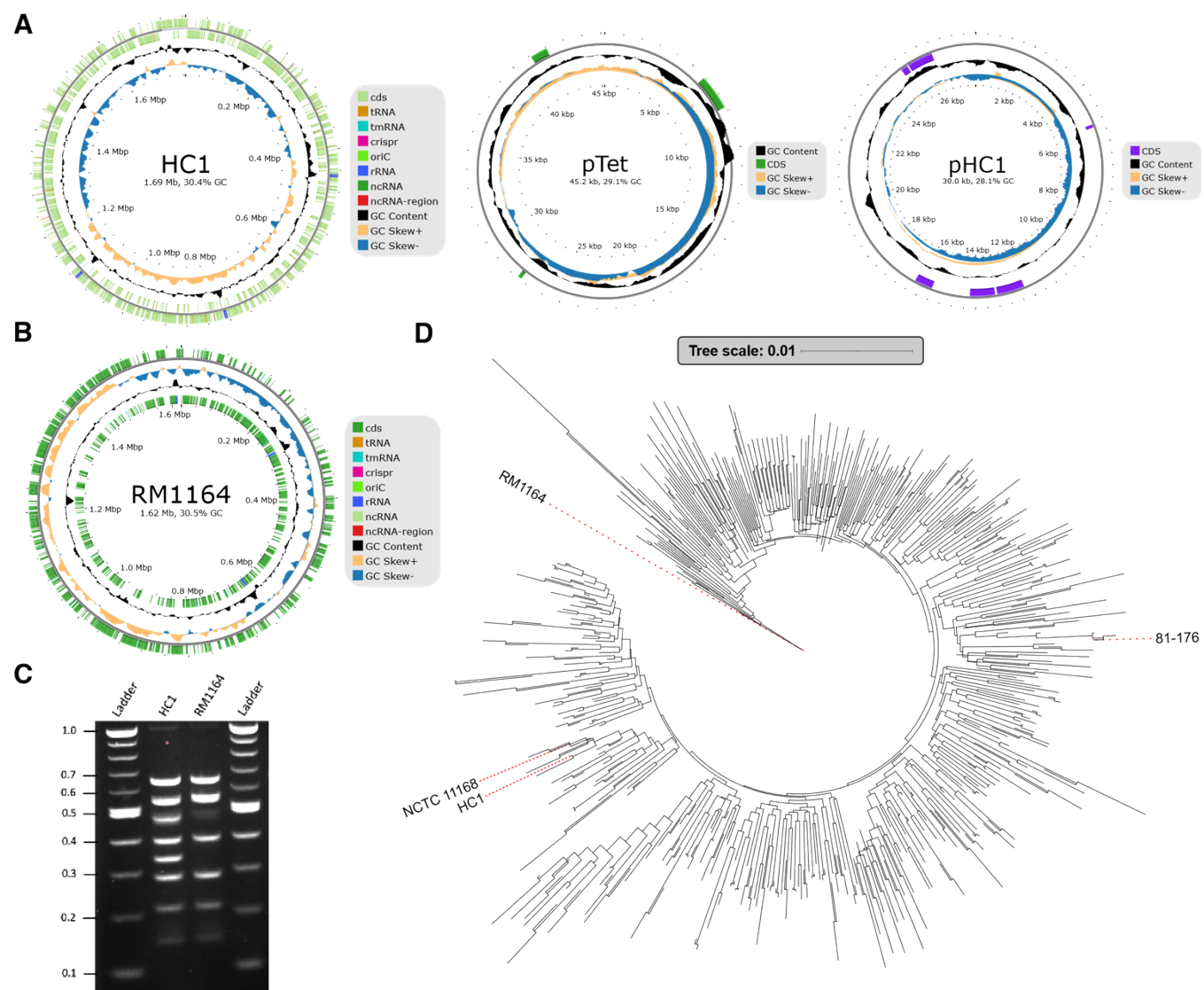
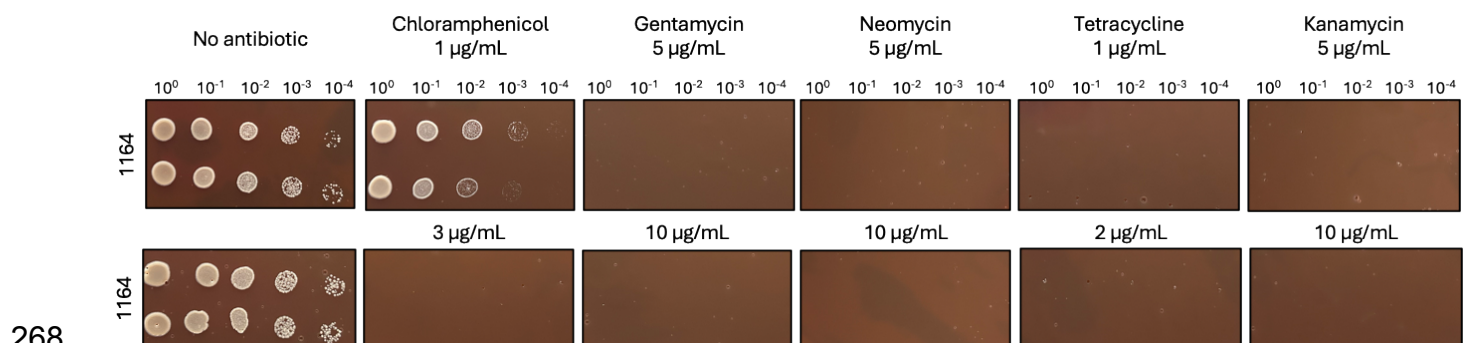


Fig 1. Whole genome sequencing of two patient and sample derived strains of *C. jejuni*.

(A) Annotated genome of *C. jejuni* strain HC1. *C. jejuni* strain HC1 also carries two cryptic plasmids, pHC1 which carries conjugative machinery and virulence factors, and pTet which carries a tetracycline resistance gene. (B) Annotated genome of *C. jejuni* strain RM1164. (C) Multiplex PCR genotyping of HC1 and RM1164. Eight primer pairs were designed from *C. jejuni* NCTC 11168 and used against HC1 and RM1164. (D) Phylogenetic relationship of the newly sequenced *C. jejuni* strains HC1 and RM1164 to 526 randomly selected publicly available *C. jejuni* genomes, as well as the reference strains NCTC 11168 and 81-176, based on whole-genome sequence similarity.

259 Antibiotic susceptibility assays

260 To identify appropriate selection conditions for *C. jejuni* RM1164, the response of the strain to a
 261 range of common selection markers was assessed via spot plating, comparing growth on
 262 selective media to non-selective growth conditions (Fig 2). *C. jejuni* RM1164 demonstrated
 263 moderate resistance to lower concentrations of chloramphenicol (2 $\mu\text{g/mL}$), however higher
 264 concentrations (3 $\mu\text{g/mL}$) were capable of suppressing growth. This strain was highly sensitive to
 265 all tested concentrations of gentamycin, neomycin, kanamycin, and tetracycline (Fig 1E). This
 266 established at least 3-4 potential selection markers capable of positive selection for each of these
 267 strains.



269 **Fig 2. Antibiotic susceptibility assays for wild type *C. jejuni* RM1164.** *C. jejuni* RM1164
 270 colonies were passaged, and spot plated onto common antibiotic selection markers. Antibiotic
 271 concentrations are measured in $\mu\text{g/mL}$. A single colony was spotted in duplicate, at dilutions from
 272 10^{-1} to 10^{-4} , and photos were obtained after 3 days.

273 Developing a shuttle plasmid for *C. jejuni*

274 We previously designed a series of multi-host plasmids carrying antibiotic selection markers, from
 275 which we designed new shuttle plasmids for *C. jejuni* [28]. One such plasmid, pHlu3, carries
 276 genes conferring resistance to chloramphenicol, and could be capable of selection in alternative
 277 hosts by cloning in *C. jejuni* codon-optimized kanamycin resistance genes. To ensure replication

in *C. jejuni*, the putative *OriC* was identified and cloned onto the plasmid, resulting in pCJ01 (Fig 3A).

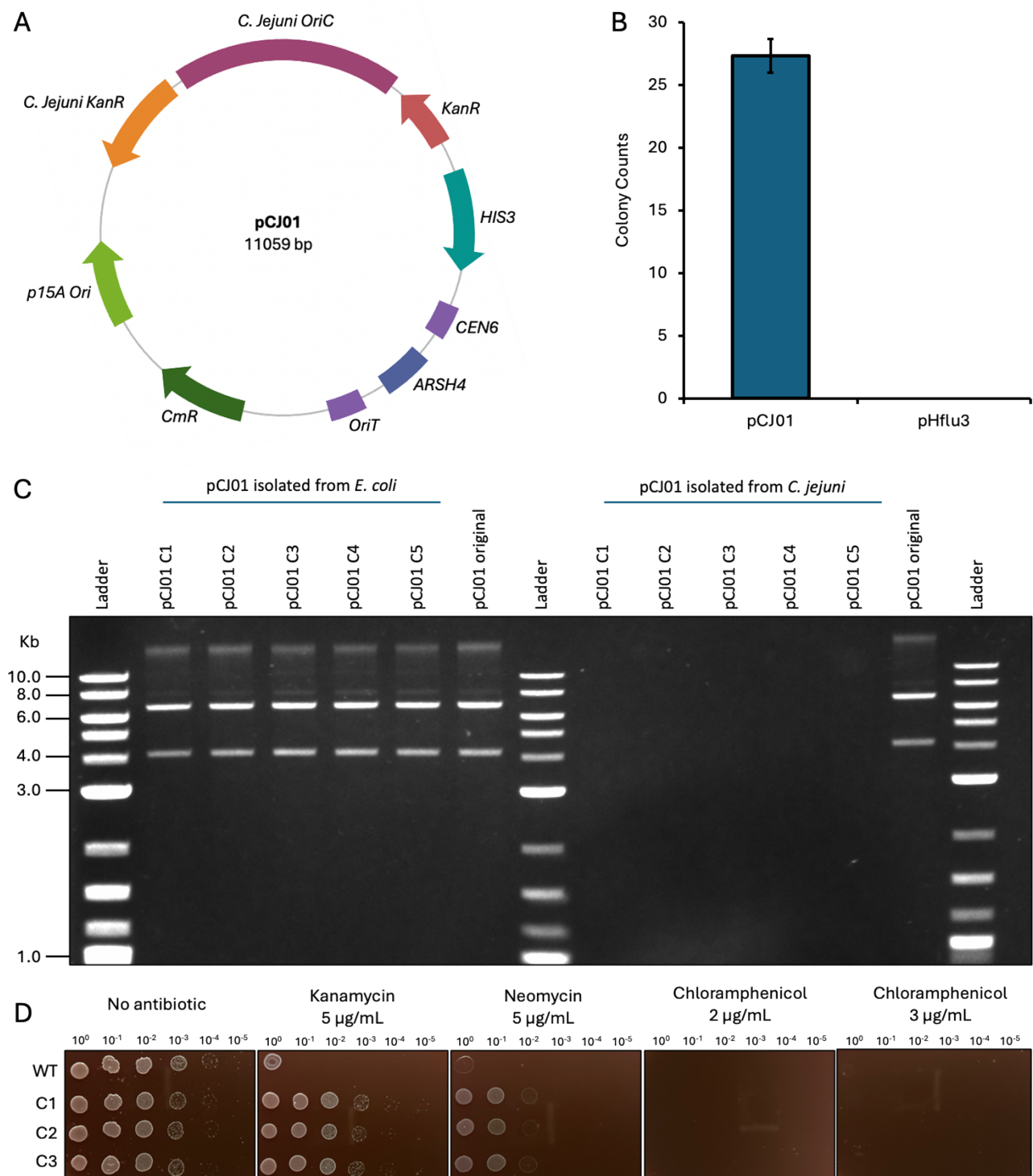


Fig 3. Developing the pCJ01 shuttle plasmid for *C. jejuni*. (A) Plasmid map for pCJ01. Kan^R, kanamycin resistance gene; p15a Ori, plasmid origin of replication; Cm^R, chloramphenicol

283 resistance gene; OriT, plasmid origin of transfer; ARSH4, yeast autonomous replicating
 284 sequence; CEN6, yeast centromere DNA element; HIS3, yeast selection marker; OriC, *C. jejuni*
 285 origin of replication. Maps were created with BioRender.com. (B) Colony counts of *C. jejuni*
 286 following transformation of pCJ01 and pHflu3 (negative control). Bars represent an average
 287 colony count from 3 independent experiments. Error bars represent the standard error from the
 288 mean. (C) Restriction digest with XbaI of pCJ01 recovered from transformants of *E. coli* and *C.*
 289 *jejuni*. pCJ01 original were the plasmids used for the *C. jejuni* transformation. (D) Antibiotic
 290 susceptibility assays for *C. jejuni* transformants harbouring pCJ01 and no plasmid *C. jejuni* (WT).
 291 Cultures were spotted at dilutions from 10^{-1} to 10^{-5} , and images were taken after 3 days.

292
 293 With the prior antibiotic susceptibility assays demonstrating sensitivity to kanamycin for RM1164,
 294 successful plasmid replication of pCJ01 in *C. jejuni* was observed via growth on selective media
 295 with kanamycin and neomycin (Fig 3B/D). To verify that resistance was conferred by
 296 transformation of pCJ01, *C. jejuni* and *E. coli* transformants were screened by restriction digest,
 297 with clear and expected band sizes appearing from the *E. coli* prep (Fig 3C). No bands could be
 298 observed from *C. jejuni* transformants, consistent with both a lower copy number of the plasmid
 299 in *C. jejuni* and poor recovery owing to insufficient domestication.

300 **Strain-specific conjugation into *C. jejuni***

301 Following the successful demonstration of pCJ01 transformation into *C. jejuni* and *E. coli*, we
 302 sought to determine if pCJ01 could be transferred from a non-pathogenic *E. coli* strain into *C.*
 303 *jejuni*. Establishing a functional bacterial conjugation system into *C. jejuni* is expected to be a
 304 tremendous step forward for *Campylobacter* genetic engineering, enabling the transfer of high
 305 molecular weight payloads beyond what is capable by standard transformation [35]. Similar
 306 advancements have been transformative for other relatively undomesticated microorganisms

[26]. As pCJ01 carries a plasmid origin of transfer (*OriT*), it can be mobilized by a helper plasmid through a process known as conjugation in *trans* [30,36]. Conjugation from an auxotrophic *E. coli* donor strain carrying the helper plasmid and either pCJ01 or pHflu3 demonstrated functional *trans*-conjugation of pCJ01 into *C. jejuni* (Fig 4A). pHflu3 can be considered a negative control due to lacking the *C. jejuni* putative *OriC* and the *C. jejuni* codon-optimized kanamycin resistance gene. To confirm resistance was conferred in *C. jejuni* by pCJ01 conjugation, seven transconjugants were isolated and PCR screened to detect for pCJ01, with all five colonies validated as successful transconjugants (Fig 4B).

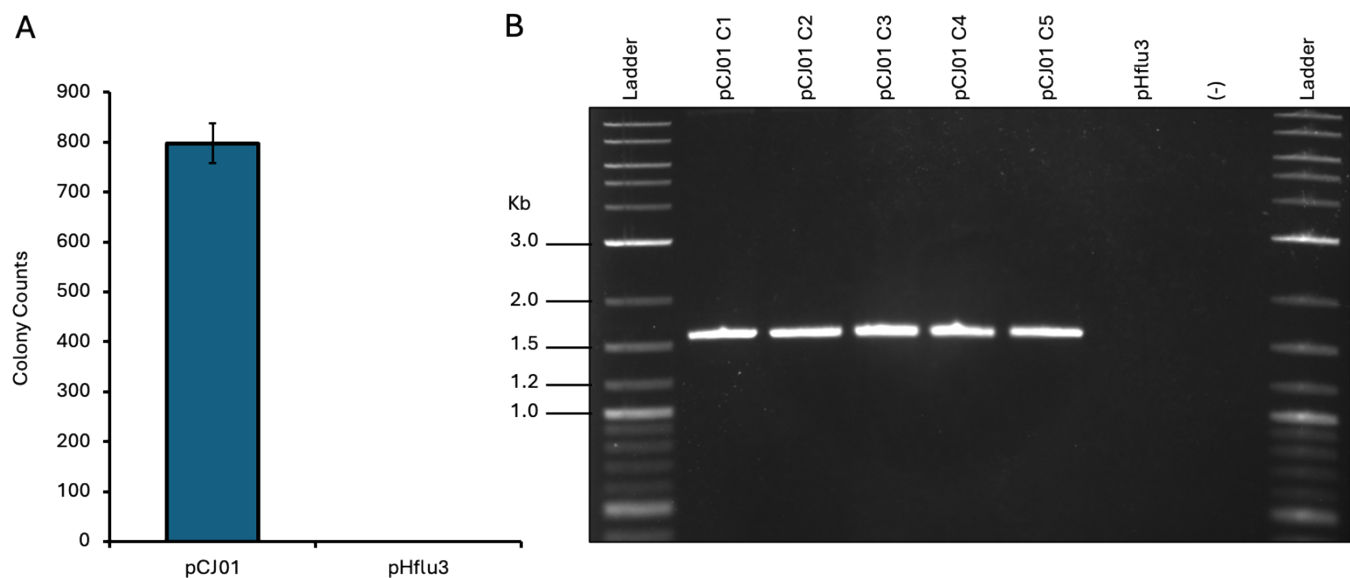


Fig 4. Demonstrating strain-specific plasmid transfer from *E. coli* to *C. jejuni* using conjugation. (A) Colony counts of *C. jejuni* following conjugation with pCJ01 or pHflu3. Bars represent an average colony count from three independent experiments. Error bars represent the standard error from the mean. (B) PCR screen of *C. jejuni* transconjugants of pCJ01. Primers specific to the *C. jejuni* Kan^R cassette produced a band of the expected size in pCJ01, but not in pHflu3. (-) represents the no-template control for the PCR reaction. The expected amplicon size is 1,678 bp.

323 Discussion

324 This work expands the genetic toolkit for *C. jejuni* by characterizing two clinically relevant,
 325 evolutionarily distinct isolates. Whole-genome sequencing revealed strain-specific differences
 326 between *C. jejuni* HC1 and RM1164, including two cryptic plasmids in *C. jejuni* HC1. However,
 327 no plasmids were detected in *C. jejuni* strain RM1164, consistent with reports that plasmid content
 328 varies widely among *C. jejuni* isolates [37–39]. In addition, including evolutionarily diverse *C.*
 329 *jejuni* strains is important because genetic tools often show strain-dependent performance and
 330 can be difficult to apply across different isolates [40,41]. We also tested antibiotic susceptibility in
 331 *C. jejuni* RM1164 and identified several selection markers that were useful for subsequent work
 332 in this strain.

333 An important result of this study is the development of a mobilizable shuttle plasmid and a
 334 conjugation mediated delivery system for *C. jejuni* RM1164. Previous genetic work in *C.*
 335 *jejuni* has relied heavily on natural transformation, which is often strain dependent [14,40,42]. In
 336 this study, we demonstrate conjugation-based plasmid delivery from *E. coli* donors into *C. jejuni*
 337 RM1164, providing an alternative route for DNA transfer. Unlike natural transformation,
 338 conjugation does not rely on the natural competence of recipient cells [43].

339 Beyond technical development, the strains and plasmids described here have practical
 340 implications for *C. jejuni* detection. *C. jejuni* and other clinically relevant strains can be difficult to
 341 culture and manipulate genetically, which has limited the design and testing of engineered
 342 biosensors for contaminated food products. A system that supports plasmid transfer and
 343 maintenance in *C. jejuni* could be useful for future development of improved detection tools for
 344 this pathogen. Current detection methods are often culture-based and can take time, which limits
 345 how quickly contamination can be identified. Making clinically relevant *C. jejuni* strains more
 346 accessible to genetic manipulation may help support the design and testing of new assay

347 strategies. However, further work is needed to quantify conjugation efficiencies across a broader
348 range of clinical isolates and to determine how reliably this system performs under conditions
349 relevant to food processing. In the longer term, tools like this could help support earlier monitoring
350 in food production settings and improve food safety.

351 **Author contributions**

352 V.N.: conceptualization, formal analysis, investigation, methodology, writing - original draft,
353 writing - review and editing; G.W.F.: formal analysis, investigation, methodology, writing - review
354 and editing; T.J.: investigation, writing - review and editing; A.E.: conceptualization, investigation;
355 M.J.R.: conceptualization, methodology, writing - review and editing; B.J.K. conceptualization,
356 formal analysis, funding acquisition, methodology, resources, supervision, writing - review and
357 editing.

358 **Competing interests statement**

359 The authors declare there are no competing interests.

360 **Data availability**

361 BAM files for whole-genome sequencing of *C. jejuni* RM1164 and HC1 were deposited in the
362 Sequence Read Archive under accession number: PRJNA1438426.

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365 the kanamycin resistance gene in this study.

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Figure 1

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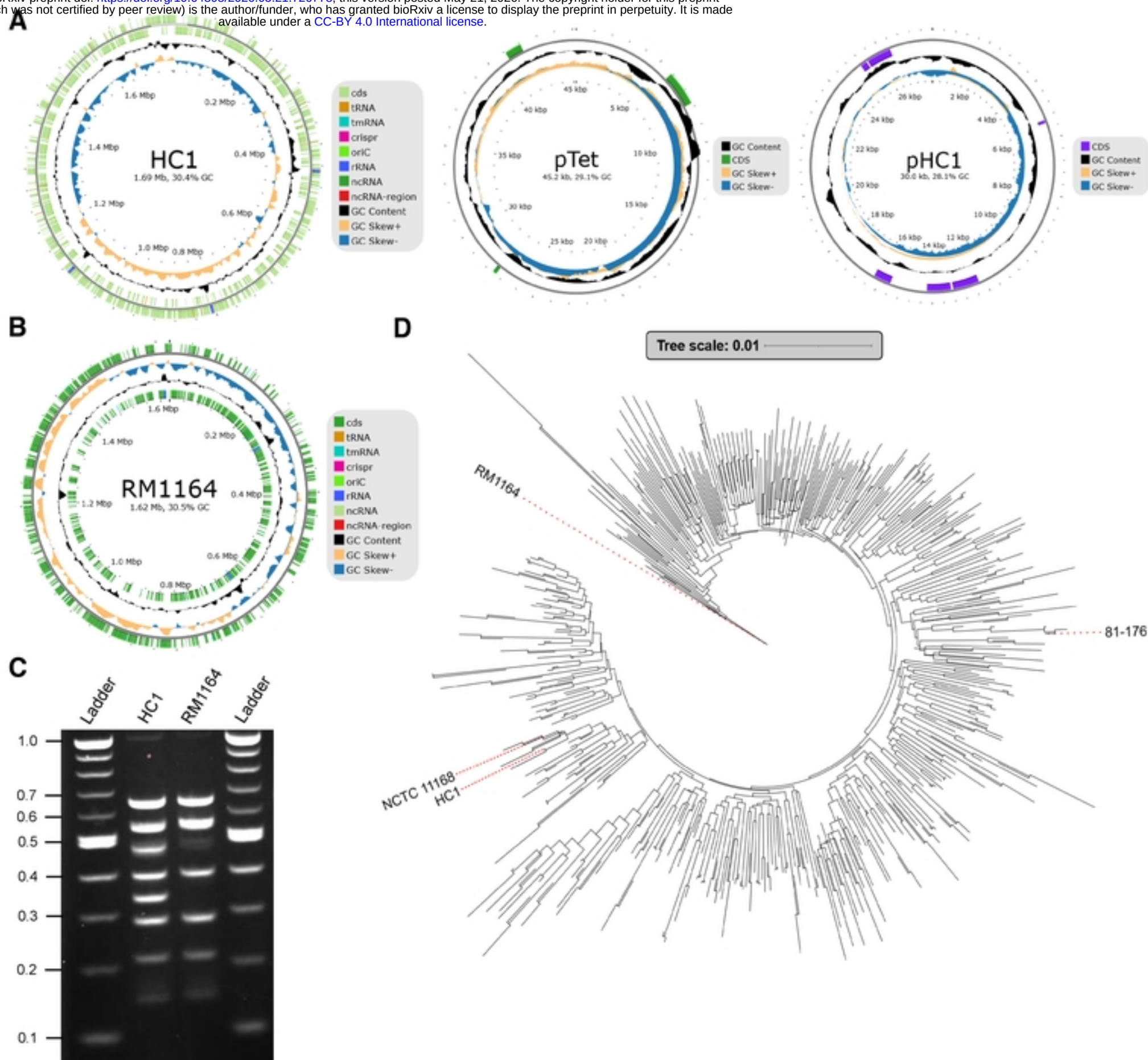


Figure 2

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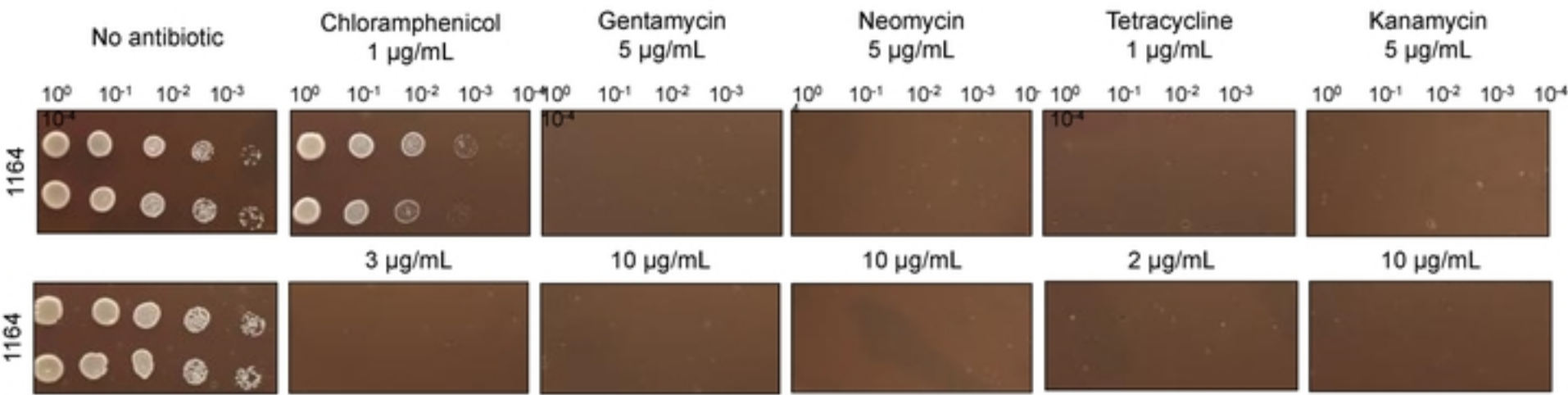


Figure 3

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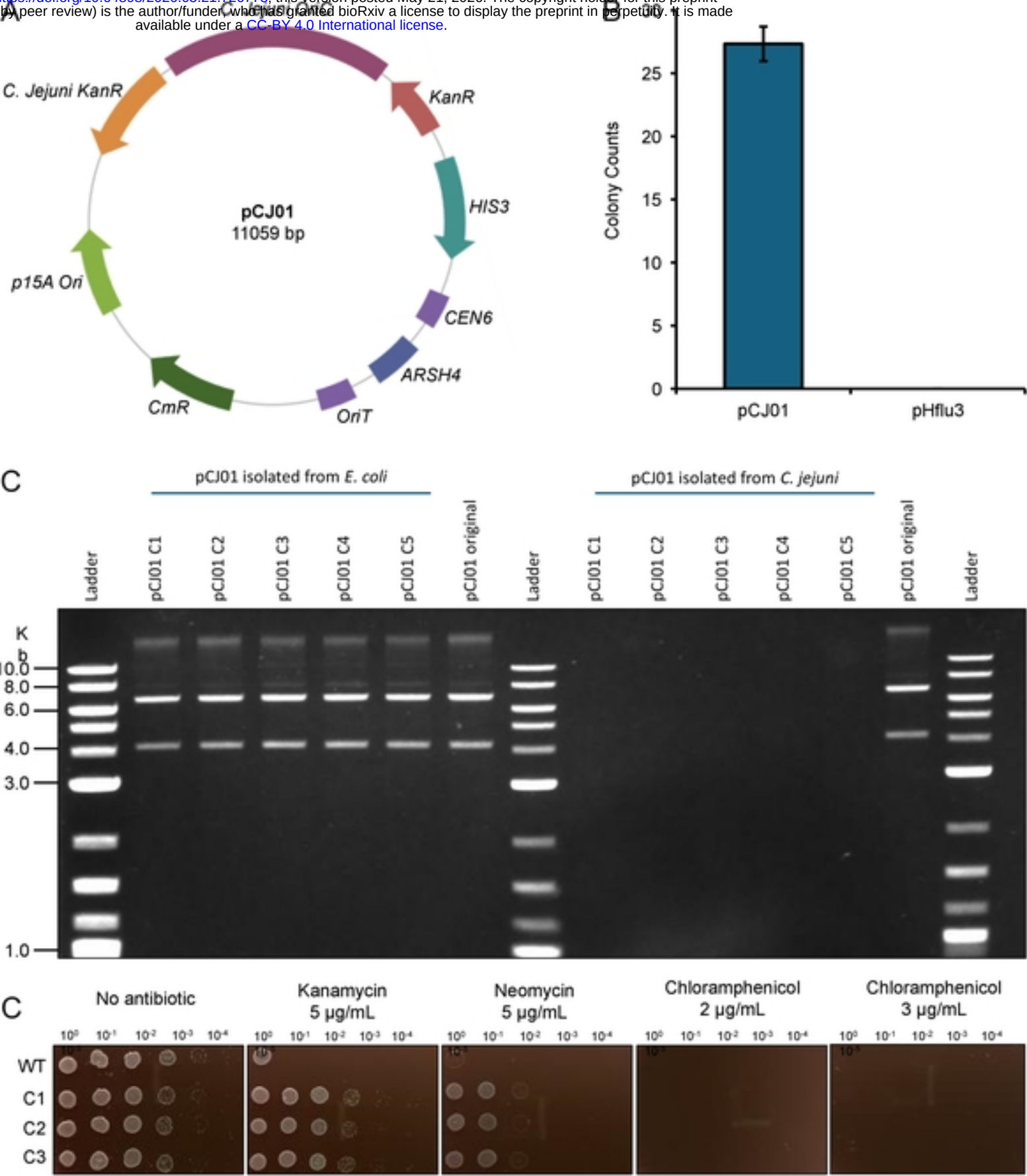


Figure 4

